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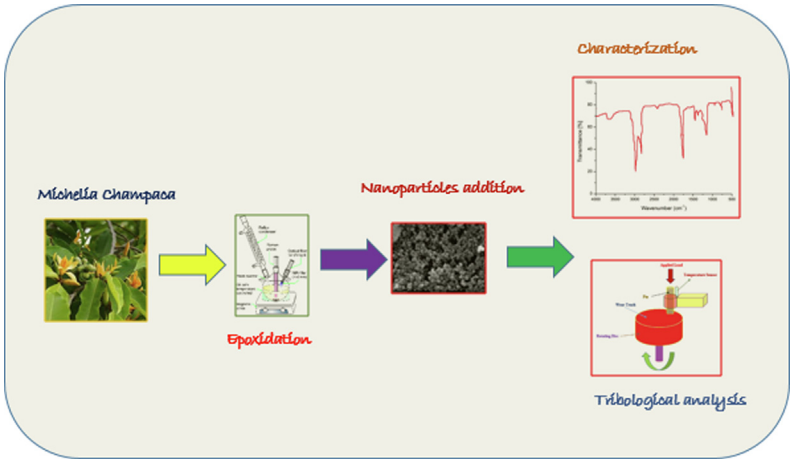
Full Length Article

Michelia Champaca: Sustainable novel non-edible oil as nano based bio-lubricant with tribological investigation

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GRAPHICAL ABSTRACT



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ABSTRACT

In this study, *Michelia Champaca* was referred as the raw oil for considering tribological investigation. The oil contains major amount of unsaturated fatty acids that was reduced by performing epoxidation process. In the investigation, CeO_2 nanoparticles were utilized as an additives to the chemically modified oil. The lubricants were examined for the thermo physical properties like kinematic viscosity, viscosity index increment, and flash point, pour point and dispersion stability with tribological analysis. Tribological characterization was conducted by using pin-on disc machine. With chemical modification of oil, an increment in the viscosity was observed. The maximum increment with reference to the viscosity was observed with 1.2% nanoparticles concentration at 100 °C temperature. The flash point increases with the addition of the nanoparticles but decrement in the pour point was observed with an increase in nanoparticles concentration. The addition of the nanoparticles to the epoxidized oil shows better dispersion stability. In this investigation, 0.6% concentration of CeO_2 nanoparticles demonstrated a reduction in the coefficient of friction (COF) and wear of the pin. The Scanning electron microscope images also show better surfaces when the nanoparticles were added up to 0.6% which was due to effective lubricant film formation at the surface. The optimum nanoparticles addition was found at 0.6% concentration to the epoxidized oil with an improved anti-wear property.

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1. Introduction

Nowadays, more focus is concentrated to explore the alternatives to the petroleum products which are environmentally friendly [1–3]. During the motion, friction is produced between the parts which promotes loss of energy. To reduce the energy loss, lubrication is required. The synthetic oils are the conventional sources which provides lubrication when the parts are in sliding motion [4,5]. The application of synthetic oils are harmful to the environment as their disposal promotes pollution. They are highly toxic and non-biodegradable. Most of the pollution created by their disposal is destructive to the water reserves. In this context, bio based lubricants are desired to meet the requirements fulfill by the conventional synthetic oils [4,6,7]. The properties which are associated for proper lubrication are viscosity, viscosity index, flash point and pour point [8,9].

Vegetable oil are the suitable substitute for the synthetic oil as they are renewable and biodegradable. They also possess properties which are important with relevance to their application as a lubricant. Vegetable oil are having some limitation that resists their application as a lubricant like low thermal stability and poor oxidation stability but it can be improved by applying certain chemical modification processes like epoxidation, trans esterification, hydrogenation etc. [10–12].

There are certain additives which could improve the properties of the vegetable oils. They are extreme pressure additives, anti-wear additives, detergents, and dispersants. The former two are the traditional additives that are mostly used. These are sulphur, phosphorous and chlorine. By forming chloride, phosphates and sulphates on the metal surfaces, protection to the applied pressure during different conditions is provided. But they are chemically harmful to the environment. The same function in the form of additives could be achieved with the application of the nanoparticles [8,9,13].

In view of literature, extreme work have been conducted with the application of nanoparticles based on their concentration, shape, and size but most of them are related to their application to the synthetic oils or edible vegetable oils. Zhang et al. [14] utilized surface grinder for the grinding application. The MoS_2 nanoparticles with 50 nm size was added and provided best lubrication to the parts. Xie et al. [15] evaluated the application of silicon dioxide and molybdenum disulphide to the engine oil tribology. The engine oil provides better tribological behavior by the application of nanoparticles. Wang *et. al.* [16] expressed the application of hexagonal boron nitride (hBN) nanoparticles to the castor oil. Improved behavior in terms of reducing friction and wear of the parts was obtained. Various metallic oxides have been used as lubricant additives. Various studies were examined for the tribological applications like ZnO [17], CuO [8], MoS_2 [15], SiO_2 [15,18], hBN [19]. Most of the studies proposed that the optimum concentration of the nanoparticles are suitable for providing better lubricity. The CeO_2 nanoparticles showed better anti-wear properties when added to the diesel engine. The CeO_2 nanoparticles is suitable as an additive to the bio based lubricants as they get easily blended with the base fluid and also provided better anti-wear properties [20].

In the previous study, none of the research was conducted while considering the application of CeO_2 nanoparticles to *Michelia Champaca* oil. Some of the studies were investigated with its application as biodiesel and performance on the engine. None of the studies quoted earlier about its application as a bio based lubricant while introducing nanoparticles.

The *Michelia Champaca* belongs to the family Magnoliaceace and particularly found in Asia. It is widely spread through India, Burma and China and is considered as a tropical plant with easy propagation. Moreover, Michelia Champaca leaves are yellowish to brown in orange in color. The seeds are dark brown in color. The leaves of this tree are utilized for the fragrance purpose in pharmaceutical industries. According to the literature, the seed of the *Michelia Champaca* plant bears excellent amount of oil that was around 44.5% [21].

The *Michelia Champaca* is the unattended non edible oil which has never been used for tribological applications. In the present investigation, raw *Michelia Champaca* oil has been considered as the reference oil and further chemical medication through epoxidation process was performed to reduce the quantity of unsaturation. In addition, CeO_2 nanoparticles are added to the modified oil to check its lubrication mechanism and improve its thermo physical characteristics. Kinematic viscosity, viscosity index, flash point, and pour point value were experimentally analyzed. The tribological characterization was performed using in on disc tribometer. A Scanning electron microscope (SEM) was used to examine the surface topography of the worn surfaces.

2. Materials and methods

2.1. Composition of Michelia Champaca

For the evaluation purpose, raw *Michelia Champaca* oil was procured from the M/s Lala Jagdish Prasad and Company, Kanpur, Uttar Pradesh, India. The oil mainly contains fatty acid triglycerides which was estimated according to European standard method EN14103.

The fatty acid composition of *Michelia Champaca* oil is analyzed by a Gas Chromatography-Mass Spectroscopy (GC–MS) (GC-10, Shimadzu, Japan) having a capillary column with 30 mm length and 0.25 mm diameter lined with a 0.25 μm (Rtx-5 ms, Rextex). Samples were injected in the split/column flow ratio 24:1. Helium was used as the carrier gas with a flow rate of 1 ml/min. The injection temperature was maintained at 250 °C and column oven also 250 °C (programmed to start at 120 °C, held at this temperature for 5 min and heated at a rate of 3 °C/min to 250 °C).

2.2. Chemical modification of the Michelia Champaca oil

The chemical modification was performed to improve the thermo physical properties of the oil. In the chemical modification process, the epoxidation process was performed. Three-necked round-bottom flask (500 ml) equipped with a magnetic stirrer was used for the epoxidation process. The central neck of the flask was connected to the reflux condenser and one neck was equipped with a temperature sensor. The remaining one side was used for pouring oil and chemicals into the system. To perform this process, 250 ml of *Michelia Champaca* oil was mixed with 24 ml of peracetic acid along with 0.3 ml of concentrated sulphuric acid as a catalyst. Afterward the solution was mixed properly using magnetic stirrer while maintaining an oil bath at 0–10 °C temperature. Around 300 rpm speed of magnetic stirrer was maintained to reach 65 °C temperature as the reaction is exothermic. After thoroughly mixing, 30 wt% hydrogen peroxide (206 ml) was poured drop wise for 30 min duration and in further processing, 500 to 700 rpm speed of magnetic was maintained for 4 h duration. After leaving the mixture for about 1 h, water was extracted from the lower surface of the burette.

For improvement of the lubricity characteristics of the vegetable oil, nanoparticles are significant enough. In boundary lubrication, nanoparticles help in building a defensive layer between the moving surfaces and also removes asperities from the surfaces. The ceria nanoparticles are considered during this investigation as they form a polishing effect and disperse in the solution in a proper way [20]. The nanoparticle are added to the modified oil on weight percentage basis (0.3, 0.6 and 1.2). The surface modifier (Triton X-100, M/s Triveni chemical, Gujarat) was used to provide stability to the nanoparticles with the epoxidized oil. The amount of Triton X-100 was 50 wt% of each nanoparticle. For

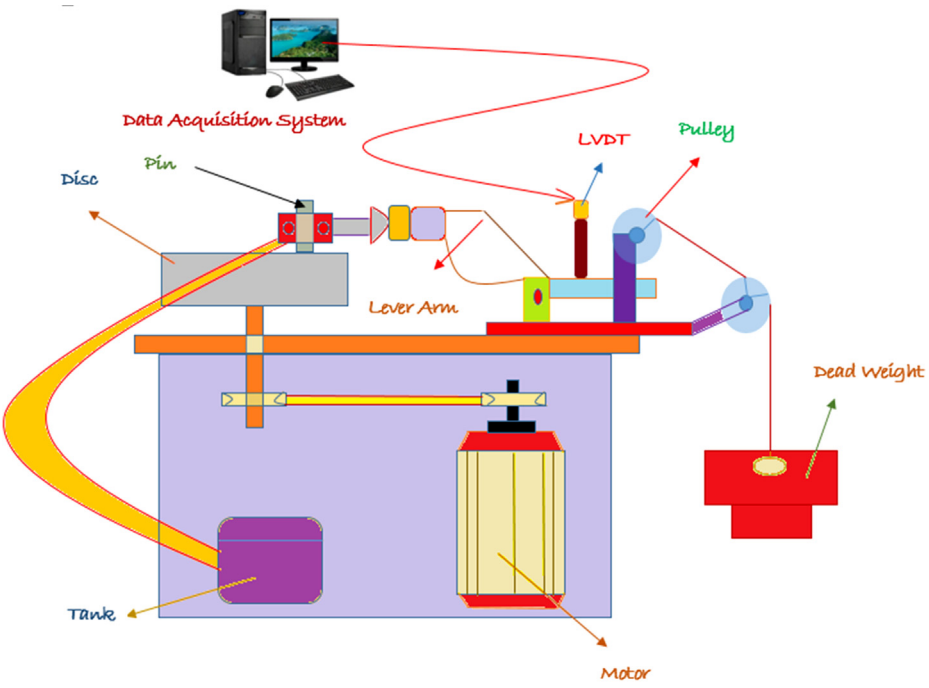
attaining steady suspension of the nanoparticles, they are mixed using a magnetic stirrer with 750 rpm/min for 3.5 h duration. A precise digital scale was used for weighing purpose of the nanoparticles and Triton X-100. Then the mixture was agitated for 45 min through the ultrasonic probe sonicator (M/s Samarth Electronics, Thane). The crystal structure of the nanoparticles were examined using X-ray diffractometer (M/s GNR India, Mumbai) as shown in Fig. 5. XRD patterns were recorded using Cu- Kα radiation (λ = 1.5406 Å) at 40 kV and 30 mA from 7° to 80° (2θ) at a rate of 3° min^{-1}. The crystallite size of CeO_2 nanoparticles were calculated based on each XRD pattern, using Debye–Scherrer's equation [17]:

$$D = \frac{0.9 * \lambda}{\beta \cos\theta} \quad (1)$$

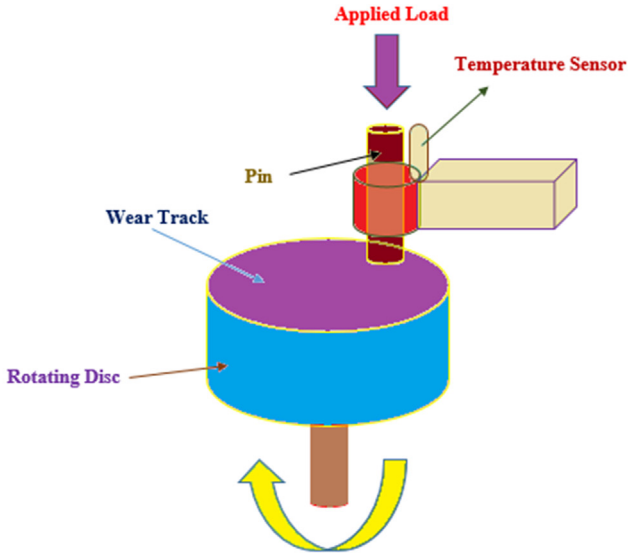
where D, λ, β and θ are the crystallites size (nm), the radiation wavelength, the full width at half of the maximum of the peak (radians) and the angular position of the peak.

2.3. Lubricant properties

The kinematic viscosity was evaluated through the viscometer (M/s Swastik systems and services, New Delhi) based on the ASTM D-445 standard. The viscosity was measured at atmospheric pressure and various temperatures (40 °C, 60 °C, 80 °C and 100 °C). The test was



(a)



(b)

Fig. 1. Schematic image of the (a) DUCOM machine (b) Mechanism.

conducted three times and the mean value was used to reduce the error and maintain reliability.

In order to calculate the viscosity index of the samples, the following equation was utilized according to the ASTM D-2270:

$$VI = \frac{[L - U]}{[L - H]} * 100 \quad (2)$$

where U represents kinematic viscosity at 40 °C, L and H are the kinematic viscosity at 40 °C with viscosity index 0 and 100, respectively, which have the same kinematic viscosity at 100 °C as the oil whose viscosity index are tried to be ascertained.

Flash point and pour point of the oil was measured according to ASTM D-92 (Cleveland open cup method) and ASTM D-97 respectively, using proper apparatus. The acid value, Iodine value and density were measured based on the standard methods published earlier [22].

2.4. FTIR spectroscopy

Fourier transform infrared (FTIR) spectrometer (M/s Bruker Alpha T) was used to analyse the disappearance of C=C bond and the formation of hydroxyl and epoxy groups. The wavenumber range was 4000–400 cm^{-1} with a resolution 32 cm^{-1}, 5 mg sample was taken in the crucible having a capacity of 50 mg.

2.5. Material for pin and disc

The LM 13 alloy (copper: 1.3%; Manganese: 0.4%; Tin: 0.1%; Titanium: 0.2%; Manganese: 1.2%; Silicon: 11.4%; Iron: 1.0%; Nickel: 1.4%; Zinc: 0.5%; and Lead: 0.1%) was procured from the (M/s Samarth metal corporation, Mumbai) which was used as the pin material for the investigation. The purpose of consideration of this material due to its application for the piston which faces maximum friction and the hardness is 98 HRB. This type of alloy is also capable to resist wear and corrosion. For the Disc, EN 31 steel was used as it contains more amount of hardness (62 HRC) and it is highly wear resistive. After purchasing ingots from the supplier, the pin with proper dimensions was made using turning operation on the Lathe machine. The pin was further cylindrically shaped. The one end of the pin was made spherical for getting the point effect while having contact with the disc. The pin consists of 12 mm diameter and length was 30 mm. The pin was further polished by using emery paper of following grit sizes 200, 400, 600 and 1200 nm before performing an experiment on the machine.

2.6. Experimental set up

The pin on disc machine was procured from the M/s DUCOM, Bangalore, India that was used for investigating the friction and wear characteristics during the application of lubricants. Fig. 1 shows the schematic image of the DUCOM machine and point contact involved in the system. The friction force was observed on the software installed in the system with the help of the load cell attached to the tribo machine. For obtaining the friction coefficient, friction force was normalized by the load applied. The lubricant was provided in a drop wise way to the interfaces with the help of the pump operated by the electric motor. The scanning electron microscope (SEM) was used to study the worn-out surfaces during the analysis. Table 1 mentioned the conditions considered for the examination of the samples.

3. Results and discussion

3.1. Fatty acid composition

The most dominant component in the raw *Michelia Champaca* oil is the linoleic acid contributing to 42.61%. The oleic acid contributed 22.71% in the fatty acid composition. Around 84% of unsaturated fatty acids are available in the raw oil which needs further chemical modification to make it a more suitable applicant for the lubrication purpose. The presence of more linoleic acid has also been reported in the work by Joshi et al [23]. Table 2 shows the amount of fatty acids present in the *Michelia Champaca* oil.

3.2. XRD characterization

Fig. 2 shows the XRD spectrum of CeO_2 nanoparticles. The characteristic peaks are located at 2θ = 28.7, 32.8, 48.4, 57.6, 57.1 and 68.9 and they correspond to (111), (200), (220), (311), (222) and (400) lattice planes, respectively. The lattice parameters from powder XRD data were found using the UNITCELL software package and the obtained values were found to be a = b = c = 5.387 Å, α = β = γ = 90°. The obtained lattice parameters for ceria nanoparticles are observed to be in good agreement with the data reported in the literature (JCPDS43-1002). The powder XRD pattern of CeO_2 nanoparticles shows broad peaks, which confirmed the formation of small-sized nanoparticles. The same diffraction pattern of CeO_2 nanoparticles has also been reported by Kashyap and Harsha with rapeseed oil [20].

3.3. FTIR analysis

Fig. 3(a) and (b) shows the FTIR spectrum evaluated for the raw oil and epoxidized *Michelia Champaca* oil. The presence of fatty acids with double bonds can be observed in the range of 3000–3010 cm^{-1} in Fig. 6(a). The C=C peak was observed at 1580 cm^{-1} wavenumber which got disappeared in the epoxidized FTIR spectrum clearly indicating its consumption. Due to the epoxidation process, evidence of the bond C–O–C was observed at 789.13 cm^{-1} in the Fig. 6(b). Similar kinds of results were also reported in the FTIR spectrum of epoxidized olive oil in the literature [24]. The presence of hydroxyl group (–OH–) in the epoxidized oil can also be observed in the range 3340–3450 cm^{-1}.

3.4. Physicochemical properties analysis

Table 3 shows the physicochemical properties of the lubricants with CeO_2 nanoparticle properties. The kinematic viscosity of raw *Michelia Champaca* oil was 73.13 mm^2/s at temperature of 40 °C and 10.11 mm^2/s at 100 °C. There was an increase in kinematic viscosity obtained with modification of the raw oil. This is due to the dissolution of the double bonds associated with raw vegetable oil [25].

The results of viscosity index (VI) indicates the temperature range that lubricant can resist during the operation. A high VI value means there is a little change of kinematic viscosity along with the change of temperature. Meanwhile, the higher value of VI ensures that lubricants samples provide good and stable lubricity through a wide range of operating temperature. According to Zulkifli et al. [26], the lubrication film becomes extremely thin at high temperatures and very thick at low temperatures. A higher viscosity index was represented by the modification of *Michelia Champaca* oil.

Table 1
Condition taken during the analysis.

Conditions considered	Value
Applied load (kgf)	12 ± 4
Sliding speed (rpm)	100 ± 20
Temperature (°C)	75 ± 3
Duration (s)	3600

Table 2
Fatty acid composition of *Michelia Champaca* oil.

Fatty acid	Molecular structure	Composition (%)
Palmitic acid	($C_{16}H_{32}$)	18.21
Palmitoleic acid	($C_{16}H_{30}$)	5.04
Stearic acid	($C_{18}H_{36}$)	2.17
Oleic acid	($C_{18}H_{34}$)	22.71
Linoleic acid	($C_{18}H_{32}$)	42.61
Linolenic acid	($C_{18}H_{30}$)	7.72

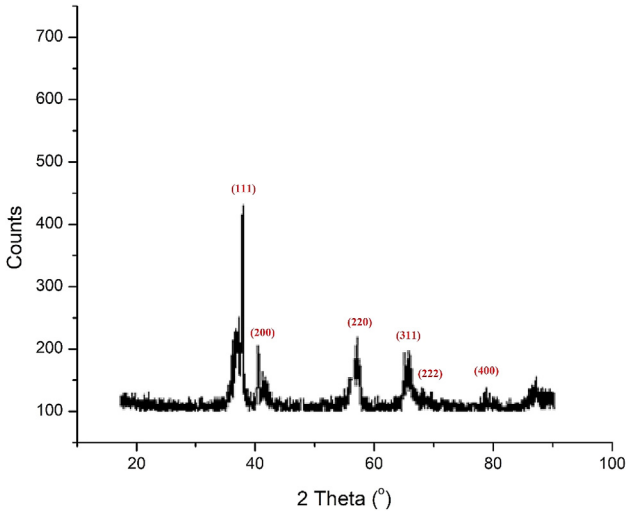


Fig. 2. XRD spectrum of CeO_2 nanoparticles.

3.5. Nanoparticles characterization

3.5.1. Viscosity increment

Fig. 4 shows the viscosity increment in percentage when the CeO_2 nanoparticles are added to the epoxidized oil in different proportions. It can be seen that the addition of the nanoparticles increases the viscosity with comparison to the epoxidized oil. The viscosity of all the lubricants decreases with an increase in temperature. The maximum increment in viscosity was found at 1.2% and 100 °C temperature. The increase of nanoparticles results in agglomeration on the surface and also molecular collision occurs which results in an increase of nano lubricant viscosity. The collision of the molecular forces with an increase in temperature reduces the viscosity of the nano lubricant at higher temperature [17]. The mathematical relation for the viscosity increment measurement is calculated by using Eq. (3).

$$Viscosityincrement\,(\%) = \frac{Nanolubricantviscosity - Epoxidizedoilviscosity}{Epoxidizedoilviscosity} * 100 \quad (3)$$

3.5.2. Relative viscosity

The relative viscosity with change in temperature for different concentrations is shown in Fig. 5. The maximum change can be observed for the 1.2% CeO_2 nanoparticles addition at maximum temperature 100 °C. The reason for the rise is already discussed in the previous sub-section. Eq. (4) shows the mathematical relation for getting relative viscosity. From the relation, it can be observed that the increment in relative viscosity is dependent on the fluid viscosity considered for the analysis. The decrease in relative viscosity with rise in temperature has also been reported by [27]

$$\mu_r = \frac{\mu_{nl}}{\mu_{eo}} \quad (4)$$

where, μ_r = relative viscosity; μ_{nl} = viscosity of nano lubricant; and

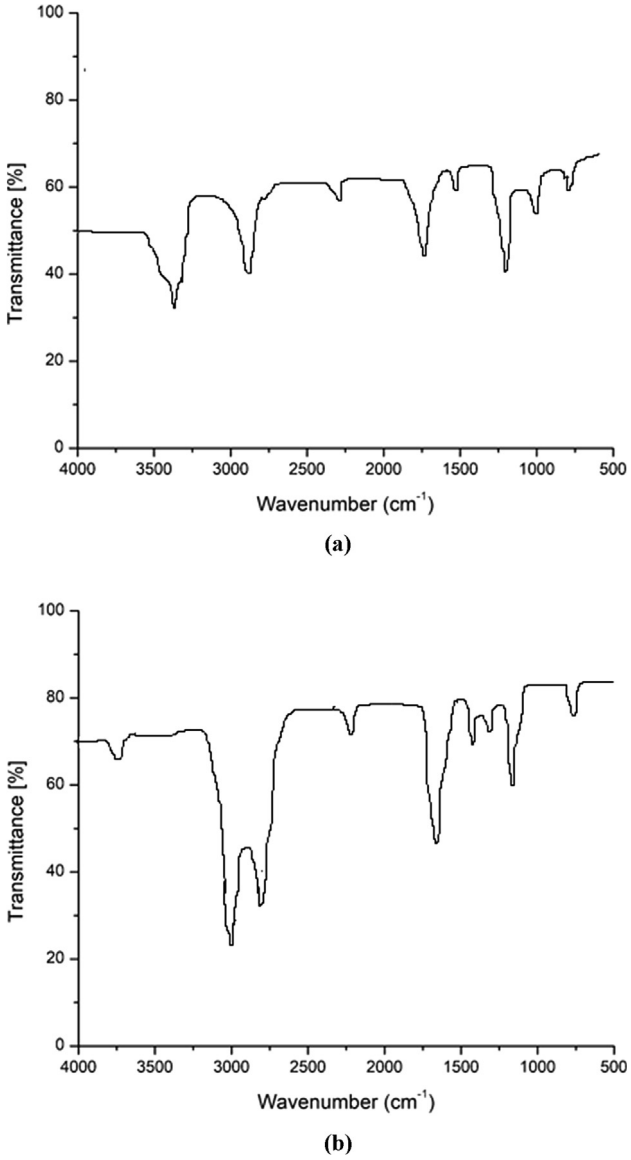


Fig. 3. FTIR spectrum of the (a) raw *Michelia Champaca* oil and (b) epoxidized *Michelia Champaca* oil.

μ_{eo} = viscosity of epoxidized oil.

3.5.3. Viscosity index increment

Viscosity index is another important parameter that decides the lubricity and it is based on the kinematic viscosity of the samples. Fig. 6 shows the variation of viscosity index increment with respect to the temperature for various concentrations of the nanoparticles. The maximum viscosity during the addition of the nanoparticles was obtained at 1.2% and at temperature 100 °C. The minimum changes in the viscosity index increment were observed at lower concentrations and temperature. These changes occurred due to higher molecular collision occurrence at a higher temperature as reported in the study of Li et al. [28].

3.5.4. Flash and pour point

The flash point is the lowest temperature at which the fluid is capable to form a flammable mixture with air and the pour point refers to the minimum temperature at which it starts flowing. Fig. 7 shows the effect of CeO_2 nanoparticles on the flash point and pour point. With an increase in the concentration of the nanoparticles, an increase in the value of flash point was observed which a good indication was. The increase in flash point provides an increased operating limit of the

Table 3
Properties of the CeO_2 nanoparticle and physicochemical properties of *Michelia Champaca* oil.

Materials properties					
Nanoparticle	Purity (%)	Size (nm)	Density (kg/m^3)	Color	Shape
CeO_2	99.9	45	7612	Brown to Yellow	Spherical
Properties	***Michelia Champaca* oil**			**Epoxidized Oil**	
Kinematic viscosity (mm^2/s) at 40 °C	73.13 ± 2.03			146.32 ± 2.14	
Kinematic viscosity (mm^2/s) at 100 °C	10.11 ± 1.01			12.14 ± 1.08	
Viscosity index	117 ± 1.07			147 ± 2.34	
Density (kg/m^3) at 15 °C	917.21 ± 1.29			889.21 ± 2.68	
Flash point (°C)	151 ± 2.44			237.13 ± 2.42	
Pour point (°C)	−4.03 ± 0.47			3.04 ± 1.06	
Iodine value	89.17 ± 2.16			11.22 ± 1.14	
Acid value (mg KOH/g)	4.56 ± 0.34			0.079 ± 0.007	

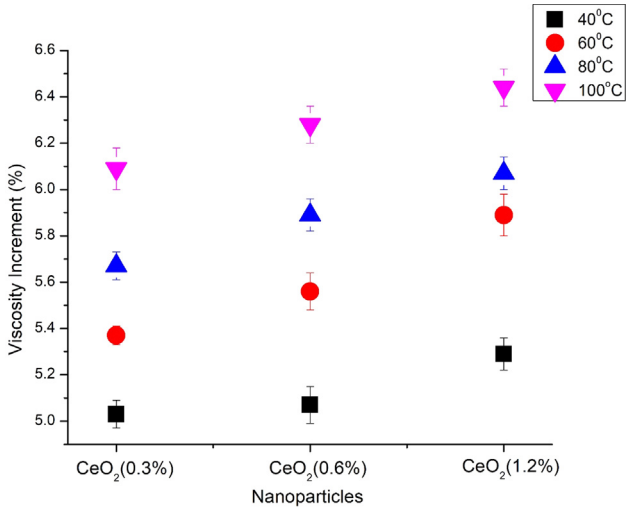


Fig. 4. Viscosity increment measurement for different CeO_2 nanoparticles concentration.

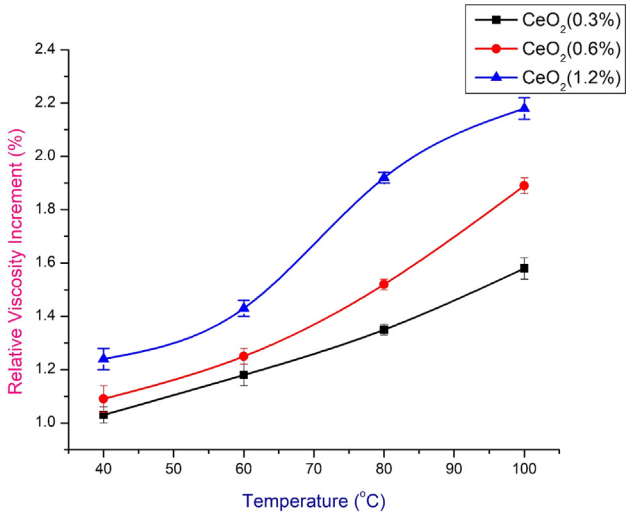


Fig. 5. Variation of relative viscosity with temperature.

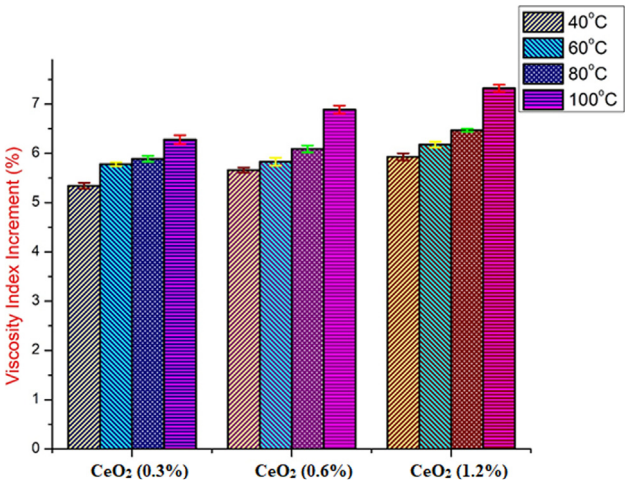


Fig. 6. Viscosity index increment with respect to the CeO_2 nanoparticles at various temperatures.

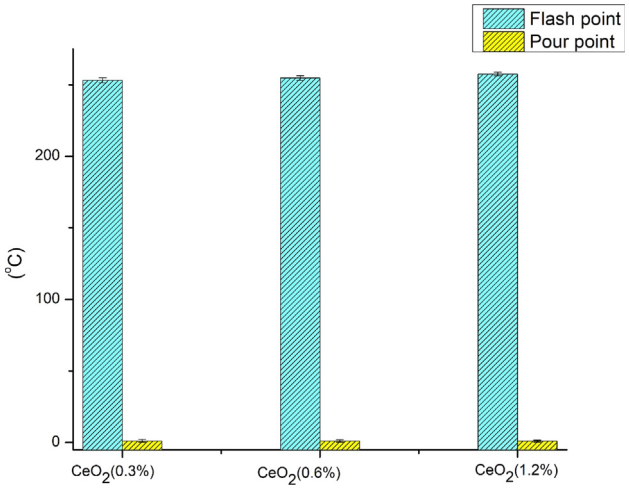


Fig. 7. Effect of nanoparticles addition on the flash point and pour point of the nano lubricant.

temperature of the lubricants. Due to an increase in the limit of operating temperature, more stability and better thermo physical properties of CeO_2 nanoparticles were observed. Hence, the enhanced flash point can be considered as an advantage in relation to enhanced lubricity characteristics of epoxidized oil. With an addition of the nanoparticles, pour point decreases. This was due to the increase in the thermo physical properties of the nano lubricants. The increase in temperature resists to the flow of the lubricant and a higher concentration of the nanoparticles doesn't provide the better flow [8,29–31].

3.6. Coefficient of friction

Fig. 8 shows the friction behavior of lubricants. It has been observed

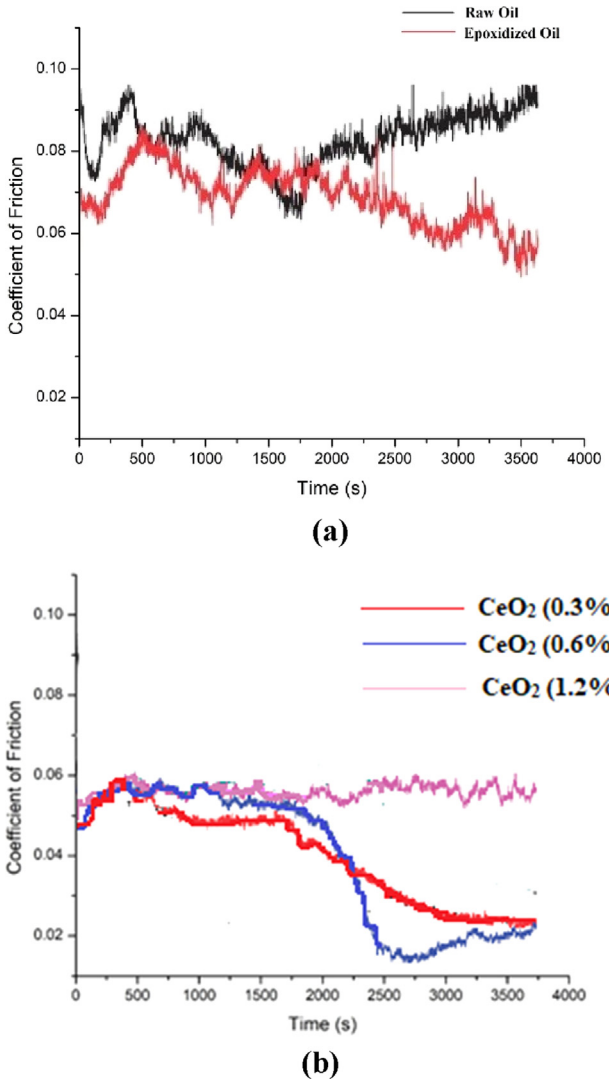


Fig. 8. Coefficient of friction with (a) raw *Michelia Champaca* oil and epoxidized oil (b) nano lubricants.

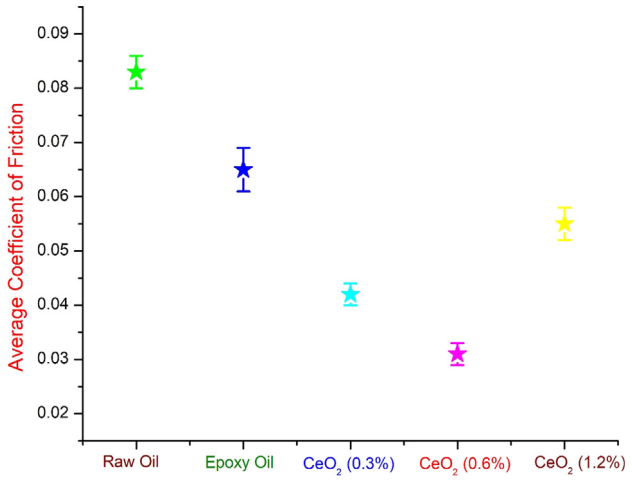


Fig. 10. Average coefficient of friction for lubricants.

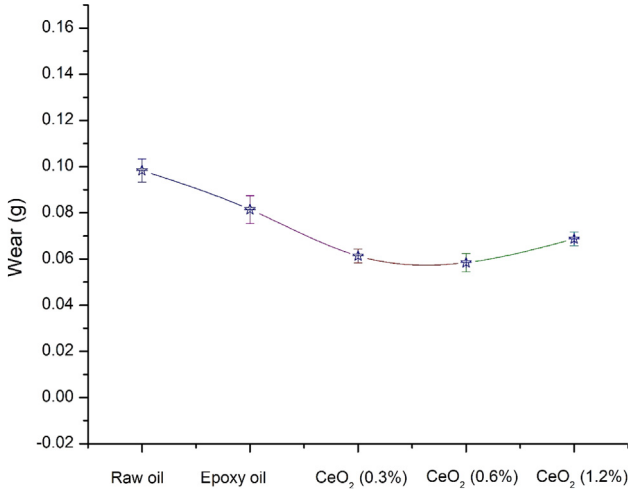


Fig. 11. Variation of nano lubricants with wear of the material.

that epoxidized *Michelia Champaca* oil presented a minimum friction coefficient with respect to the raw oil as shown in Fig. 8(a). This was due to the formation of a thick film that was created by the tribo-chemical reaction. It has been reported that the formation of the

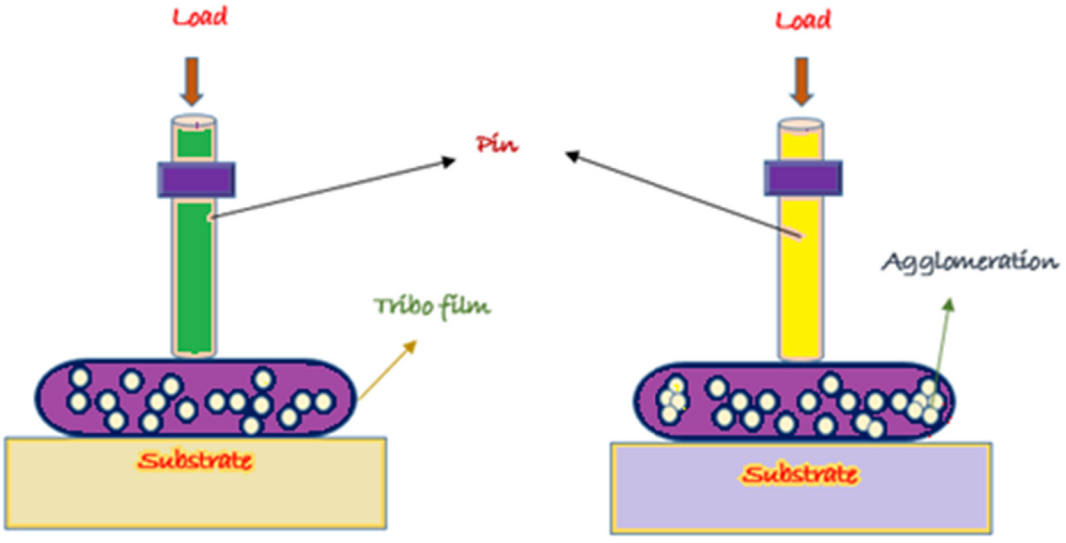


Fig. 9. Schematic description of the nanoparticles concentration-effect showing (a) effective lubrication (b) high agglomeration of the nanoparticles.

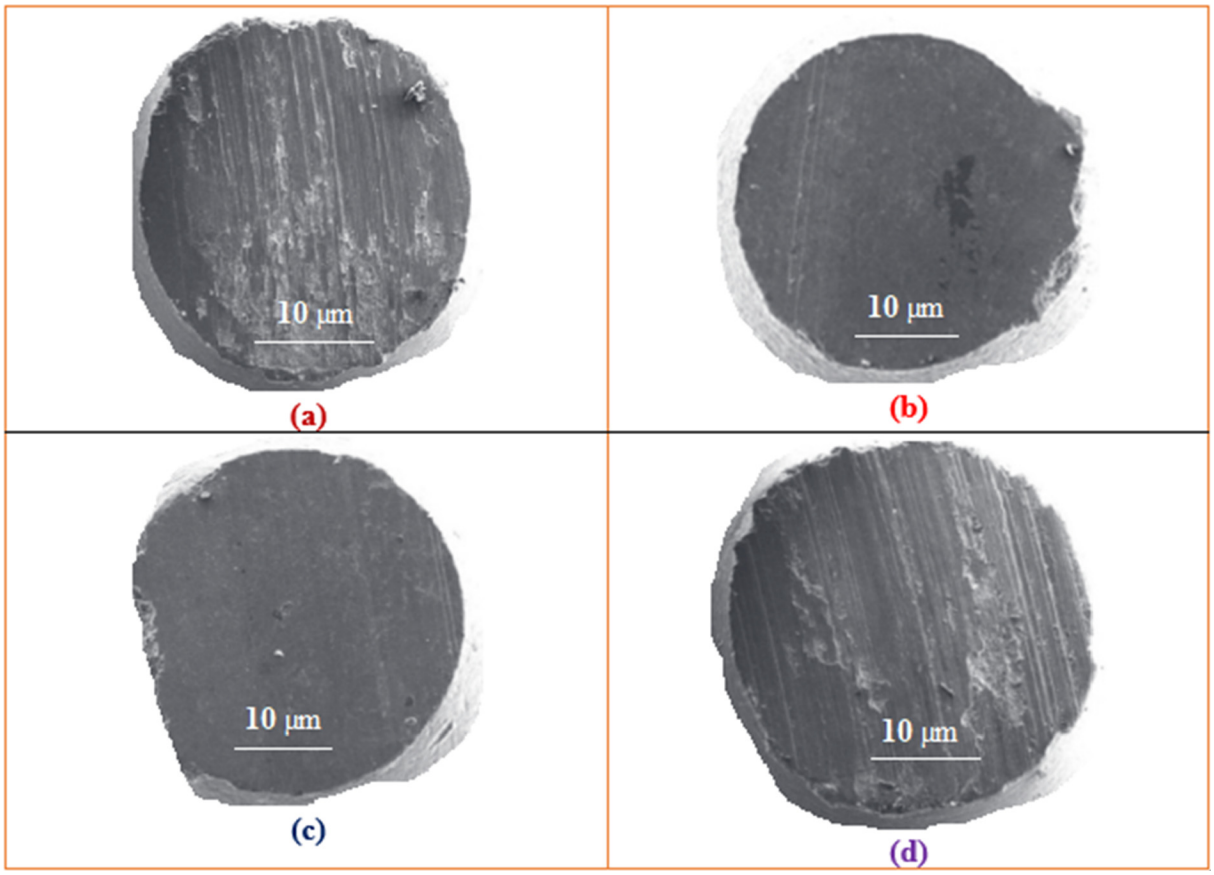


Fig. 12. SEM images of the lubricants for (a) Raw *Michelia Champaca* oil (b) Epoxidized *Michelia Champaca* oil (c) CeO_2 (0.6%) (d) CeO_2 (1.2%).

oxirane ring provides an effective layer on the surfaces during the motion [32]. The raw oil initially shows the maximum coefficient of friction which showed increased behavior and takes time to get stable on the surface.

The effect of nanoparticles addition is depicted in Fig. 8(b). The minimum coefficient of friction was obtained when the concentration of 0.6% nanoparticles are there in the chemically modified oil. The coefficient of friction was reduced around 16.22% when compared to the raw oil. The mechanism of friction reduction involves the filling of gaps on the surfaces with nanoparticles which provides smoother and better lubricity. With the addition of the nanoparticles, contact surface area increased which resulted in the reduction of the applied pressure and rolling mechanism occurs instead of sliding friction. The same reason has also been reported by Kotia et al. [33].

Based on the above results, it can be inferred that an optimum concentration of the nanoparticles is desired to receive better tribomechanism. With an increase in the amount of nanoparticles after a limit, particles get aggregated on the metals which enhances friction and wear on the surface during their contact. Fig. 9 shows the tribomechanism involved while considering the proportions of the nanoparticles. It can be observed that there is a requirement of optimum concentration of the nanoparticles to maintain proper lubricant film on the surface. Higher concentration in the nanoparticles produces abrasive behavior which results in a reduction of the nanoparticles due to resistance to the flow of the lubricant. The same hypothesis has also been reported in the previous studies and also presenting abrasive wear with an increased nanoparticles concentration [8,34,35]. Fig. 10 shows the average coefficient of friction for nano lubricants. Reduction in the amount of coefficient of friction was obtained during 0.6% nanoparticles concentration.

3.7. Wear

To examine the wear of the pin, the samples are properly washed with acetone and dried in a hot oven before weighing on the machine. The wear of the pin was minimum for the epoxidized *Michelia Champaca* oil with comparison to the base oil as evident from Fig. 11. The epoxidized oil forms an improved lubricant layer between the surfaces during their contact. The epoxidized oil contains polar groups that promote adsorption on the surface resulting in minimum wear [20].

The concentration of CeO_2 nanoparticles up to 0.6% showed minimum wear of the pin when compared to the raw *Michelia Champaca* oil. The reduction in the wear was around 3.17% with the introduction of 0.6% nanoparticles. The nanoparticles effect on the anti-wear property depends on the proper dispersion of the particles in the epoxidized oil. The CeO_2 nanoparticles are having the better capability to get properly dispersed in the solution which promotes anti-wear mechanism.

3.8. Worn surfaces

Fig. 12 shows the SEM images of the lubricant samples obtained for the worn surface study. Fig. 12(a) shows the SEM images for the base oil. The pitting and severe grooves was observed on the surface due to break down of the lubricant film. For epoxidized *Michelia Champaca* oil, as depicted in Fig. 12(b), the surface was found smoother and there is no delamination of the layers. This was due to the formation of -O- cross-linking on the surface which protects it by assisting the lubricant film [36–38]. The addition of the nanoparticles results in less wear as they provide a defensive film between the surfaces during their motion. Fig. 12(c) shows the SEM images with the effect of 0.6% copper oxide nanoparticles addition. The minimum damage to the surface was

observed which was due to the proper filling of the asperities present on the surfaces and provide resistance to the applied pressure. However, a further increase in nanoparticles results in more amount of wear volume. Fig. 12(d) shows the effect of adhesion and abrasion on the surface when the amount of cerium oxide nanoparticles further increased. This was due to the agglomeration of the nanoparticles on the surface and results in the wear of the surface [31].

4. Conclusions

The *Michelia Champaca* oil was chemically modified and the effect of cerium oxide nanoparticles at various concentrations was evaluated for the tribological applications. Based on the observations, improved behavior by the application of nanoparticles to the epoxidized oil is achieved. The following observations are drawn from this study:

- With the addition of the nanoparticles, an increase in the viscosity index percentage was observed. A higher amount of increment was obtained at higher concentrations.
- The addition of the nanoparticles to the chemically modified oil increases the flash point of the nano lubricants. A higher value of the flash point was obtained at 1.2% which was around 2.01% increment from the epoxy oil.
- Adding nanoparticles to the fluid decreases the pour point. There are marginal changes in the pour point at lower nanoparticles concentration i.e. 0.3% and 0.6%.
- The epoxidized oil shows improved lubricity in comparison to the raw *Michelia Champaca* oil in terms of reducing friction. This was due to the presence of –O– cross-linking in the surfaces which successfully forms a protective film on the surface.
- The addition of the CeO_2 nanoparticles to the chemically modified oil improve its lubrication behavior. With the addition of up to 0.6% concentration, better results are obtained in terms of reducing friction. Their addition to the lubricant assists in forming a better defensive film between the surfaces and prevents direct contact of metals during the motion.
- The raw oil shows maximum scratches on the surface with comparison to other lubricants. This was due to the delamination and scratches on the surface. The epoxidized oil shows less wear to the surface with comparison to the raw oil as there is presence of double bonds in the molecular structure. The addition of the nanoparticles up to 0.6% to the modified oil shows less wear and damage to the surface.

CRediT authorship contribution statement

Yashvir Singh: Writing - original draft. **Erween Abd Rahim:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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